

Abstract No.

Acquired genetic and cell state changes in IDH-mutant glioma progression

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Astrocytomas and oligodendrogliomas exhibit variable treatment response and patient outcomes. To



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better understand the genetic and cellular changes that drive IDH-mutant glioma progression, we analyzed longitudinally collected whole-genome/whole-exome datasets from 183 IDH-mutant glioma patients available through the Glioma Longitudinal Analysis (GLASS) Consortium, combined with single-nucleus transcriptomic profiling of 75 temporally separated tumors across 35 oligodendroglioma/astrocytoma patients. At recurrence, we observed enrichment of hypermutation across glioma types, astrocytoma-specific *CDKN2A/B* homozygous deletion and focal oncogene amplifications, and oligodendroglioma-specific PI3K/NOTCH point mutations. Astrocytomas but not oligodendrogliomas often additionally acquired new chromosomal imbalances, which converged with chromosomal instability markers such as chromothripsis, whole-genome doubling and radiation-treatment associated ID8. Non-hypermutated temozolomide-treated oligodendrogliomas showed a significant enrichment in SBS119, indicating a distinct mechanism for resolving temozolomide-induced damage. These findings show that the genomes of oligodendrogliomas and astrocytomas adapt to the selective pressures of tumor progression and treatment in distinct manners. We next asked whether glioma progression may be associated with changes in cell states. We found that malignant cell states across both tumor types transcriptionally resemble stages of normal glial-neuronal lineage development or a reactive mesenchymal-like state. Longitudinal analysis highlighted similar malignant cell state transition patterns in astrocytomas and oligodendrogliomas. Specifically, tumors that acquired recurrence-associated genetic events, including treatment-associated hypermutation, increased copy number alterations, and cell cycle alterations, showed reduced lineage differentiation and increased proliferative malignant cells at recurrence. An increase in mesenchymal-like state abundance occurred independently of acquired genetic alterations, while no other cell state changes were observed in tumors with relatively stable genomes. Thus, despite clear differences in how astrocytoma and oligodendroglioma genomes adapt to treatment-induced selective pressures, both tumor types converge on shared transcriptionally-defined malignant cell states that associate with reduced overall survival. Our results show that oligodendrogliomas and astrocytomas diverge in genetic response to treatment but converge on cell state adaptations.